

DANIEL RIBES

Senior Health Economist · HEOR & Market Access · AI / GenAI Engineer

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PROFESSIONAL SUMMARY

Senior Health Economist and AI engineer with 10+ years bridging rigorous HEOR methodology, advanced analytics and software engineering. Proven record leading HTA submissions (NICE, NCPE, SMC) and building complex economic models — cost-effectiveness (CEA), budget impact (BIM), partitioned-survival, Markov and dynamic-transmission — across oncology, rare disease, cardiometabolic and renal conditions. Over the last year I have delivered a portfolio of freelance HEOR models alongside production-grade generative-AI systems (retrieval-augmented evidence agents, multi-model LLM orchestration, and automated evidence-to-deliverable pipelines) and completed an MSc in Data Science & AI with Distinction (Oct 2026).

TECHNICAL SKILLS & CORE COMPETENCIES

Health Economics & HTA: CEA, Budget Impact Models, Partitioned Survival, Markov / semi-Markov, Dynamic Transmission, Microsimulation, PSA / OWSA / DSA, survival & evidence synthesis (ITC/NMA, MAIC, Bucher, STC), Global Value Dossiers, model QC & validation.

Programming & Data: Python (pandas, Streamlit, python-pptx/docx), R (ODEs, survival), VBA & Excel COM automation, scikit-learn, PyTorch, SHAP, FastAPI, Docker / GitHub Actions, Hugging Face Spaces, Power BI / Fabric, C++, Julia, SQL, TypeScript / Next.js, WinBUGS.

AI & GenAI: Retrieval-Augmented Generation (ChromaDB, sentence-transformers), LLM agents (LangGraph), multi-model orchestration & MCP server development, prompt engineering, NLP (transformers, topic modelling, sentiment), OpenAI Whisper, IBM Docling; Claude / OpenAI / Gemini APIs; agentic coding (Claude Code) and human-in-the-loop AI model-QC benchmarking.

Blockchain & DevOps: Solidity, Hyperledger, full-stack development, Git, CI/CD.

Therapeutic Areas: Oncology (prostate, RCC, NSCLC, lymphoma, NET, CAR-T), rare disease, cardiometabolic (obesity, diabetes, Lp(a) / CV), renal, vaccines (RSV).

PROFESSIONAL EXPERIENCE

Freelance HEOR Modeler & AI Engineer | DANRIBES S.L. — Barcelona (Remote)

Nov 2025 – Present

Freelance portfolio of economic models and GenAI systems (engagements anonymised by area):

- **Cost-effectiveness modelling** — built an oncology CAR-T model with cure-fraction dynamics, life-table background mortality and empirical Kaplan–Meier survival; engineered a live-recalculation probabilistic sensitivity analysis (PSA) engine, OWSA tornado and CEAC in Excel/VBA, plus client cure-dynamics decks.
- **Budget impact modelling** — delivered a large multi-line (1L/2L/3L) lymphoma model with time-phased cohort transitions, a market-share ramp engine, ASP/WAC price toggling and a navigable “staircase” dashboard architecture; reconciled net budget impact across regimens; authored the AMCP-format Technical Report and a full input reference verification (774 inputs traced to the source page across 182 highlighted documents, live conversion formulas, prioritised action list) that surfaced engine double-inflation and trial-arm mismatches.
- **Health economic evaluation & QC** — developed a renal Phase 2 economic evaluation with DRG-based and Medicare (CMS) costing modes; ran independent QC audits (incl. a cardiovascular Lp(a) combined model), resolving hundreds of validation errors to zero; independent formula / VBA audits of client models and a critique of an AI model-QC platform test plan.
- **Methodology & applied research** — designed an AI-vs-rule-based model-QC benchmark — a 100-error fault library across 72 seeded models with a VBA detection engine — quantifying where LLM reviewers match or miss deterministic checks; authored ITC/NMA feasibility tooling (MAIC, Bucher, STC, Schoenfeld power); ISPOR 2026 abstract and poster prepared on a pre-registered, human-in-the-loop AI QC workflow for HEOR Excel models.
- **GenAI engineering** — built retrieval-augmented (RAG) evidence agents (LangGraph + ChromaDB + Streamlit) answering clinical/HEOR questions with source attribution; automated evidence-to-deliverable pipelines turning Excel results and reference libraries into branded decks and reports; developed and integrated multiple Model Context Protocol (MCP) servers for multi-model LLM orchestration and a context-compression proxy for long-context workflows; designed HEOR Studio (AI-native toolkit of skills, MCP servers and workflows that builds documented R / Excel models) and an HEOR Report Hub.
- **Evidence synthesis & client communication** — targeted literature review protocol and extraction for a rare liver disease (HDV) programme with corpus-verified claims; MLR-ready infographics and a cost-calculator model for a renal product;

value-proposition, AMCP-dossier and CEM / BIM scoping for NSCLC and IBD assets; webinar on AI in the EU Joint Clinical Assessment.

Engagement Manager | IQVIA — Barcelona (Remote)

Jul 2025 – Oct 2025

- Led projects leveraging the IQVIA Core Obesity Model and integrated Core Obesity–Diabetes Model, demonstrating treatment value across 20+ year cardiometabolic horizons.
- Drove client partnerships and coordinated teams to develop patient-level, peer-reviewed integrated disease models (diabetes, obesity, cardiovascular disease, CKD, cancer), ensuring timely, high-quality delivery.

HTA Modeler & AI Integration Project Lead | AXIS Consulting UK — Barcelona (Remote)

Oct 2024 – Jul 2025

- Coordinated HTA submissions to NICE, NCPE and SMC for clients including Beigene, Pharmanovia and UCB; developed a viability decision-making model for a NICE assessment.
- Integrated AI solutions to streamline consultancy workflows, significantly reducing manual processing time in evidence generation while improving insight quality.

Lead Modeler — Arexvy RSV Value & Evidence Team | GSK (Freelance) — Barcelona (Remote)

Jul 2021 – Oct 2024

- Led development of a hybrid Excel / R / C++ ODE-based dynamic-transmission cost-effectiveness model for Arexvy (RSV vaccine), informing global vaccination policy (collaboration with ICON PLC and RTI).
- Designed a scalable multi-country model and built static CEMs and Budget Impact Models with flexible architectures for market localisation (collaboration with CHES).

Region Europe HEOR Manager | Novartis Oncology — Basel, Switzerland

Jun 2020 – Jun 2021

- Developed a matrix engine for Neuroendocrine Tumours (NETs) enabling flexible pricing and volume-capping across multiple products; established a HEOR Centre of Excellence.
- Managed pricing and commercial constructs for Sickle Cell Disease therapies tailored to the UK market.

Health Economist Modeler — Head | Wellmera AG — Basel, Switzerland

Mar 2018 – May 2020

- Built a 262-state non-homogeneous semi-Markov cohort model for Cushing's Syndrome (Osilodrostat) with an executive UI for non-technical stakeholders.
- Published research in Kidney International Reports demonstrating the economic value of Acute Kidney Injury (AKI) interventions via decision-tree modelling.

Earlier: Health Economist Modeler, Envision Pharma–Curo (UK, 2016–2018) · Health Economist Modeler, Wickenstones Ltd (UK, 2015–2016) · Health Economics Intern, Novartis Oncology (UK, 2015).

SELECTED TECHNICAL PROJECTS — [GITHUB.COM/DANRIBES](https://github.com/danribes)

- España en escenarios (TFM, MSc Data Science & AI, Evolve, defended Sep 2026) — transparent macro-fiscal simulation and AI-model evaluation platform for Spain: semi-structural scenario engine (Python / TypeScript) with stochastic debt projections, regional housing models with bootstrap uncertainty, a gradient-boosting sovereign-default classifier with SHAP, and a RAG assistant answering with cited passages; dated-cut-off data provenance and reproducible builds; deployed on GitHub Pages and Hugging Face (github.com/danribes/tfm-data-science).
- Hypertension microsimulation model (Python) for aldosterone-synthase inhibitors — dual-branch phenotyping and atrial-fibrillation tracking.
- RenalGuard AI (TypeScript) — CKD management platform assisting primary-care physicians with AI-driven patient management.
- Interactive Budget Impact dashboard (Python / Streamlit) and reusable AI agent “skills” for HEOR deck, report and protocol generation.

EDUCATION & CERTIFICATIONS

MSc Data Science & AI (with Distinction) — Evolve Academy

Oct 2026

MSc Blockchain Development (Full Stack, Solidity, Hyperledger) — Codecrypto Academy, Madrid

2024

Advanced Modelling Methods for Economic Evaluation — Centre for Health Economics, University of York

2017

MSc Health Economics (with Distinction) — City, University of London

2015

Management Development Program — IESE Business School, Univ. de Navarra

2005–06

MSc Agricultural Engineering — UPC (Barcelona) & Wageningen University

1993

LANGUAGES

Spanish, Catalan (native) · English (fluent) · French, Italian (advanced) · Portuguese (intermediate) · German (basic)